Business Intelligence System for UA

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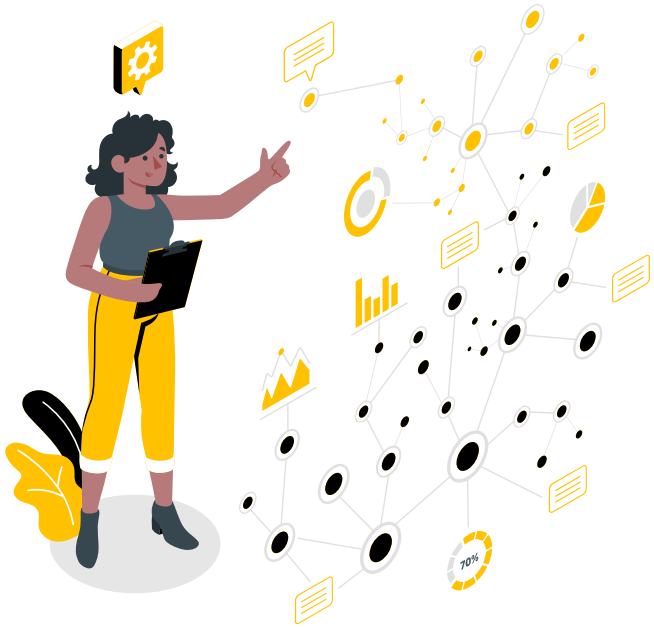
Problem



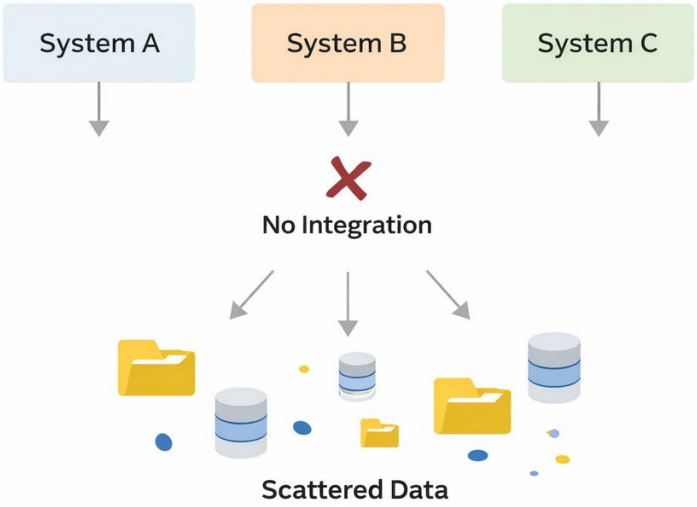
01

Challenges caused by fragmented data and disconnected systems

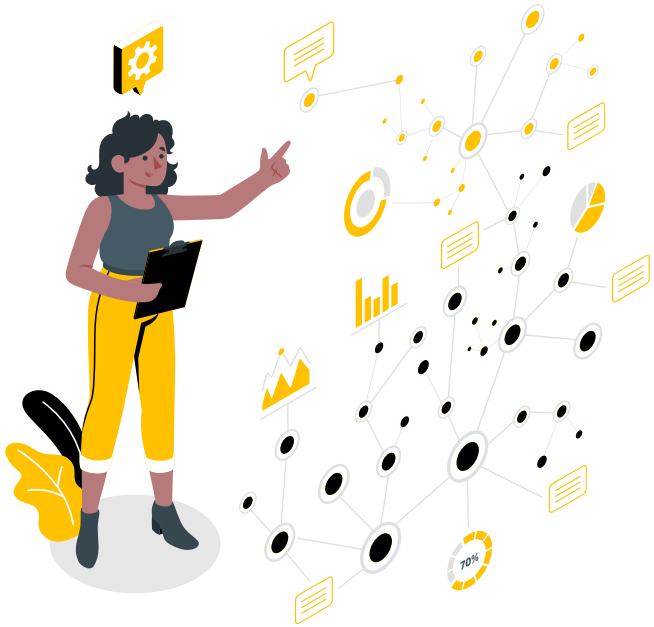
Problem

Key Problems in Organizations:

- Multiple independent systems;
- Data scattered across platforms;
- Lack of integration between systems;
- No unified view of the organization;
- Difficult and slow decision-making.



Context



02

Overview of UA systems and current data landscape

UA Context

Systems

Independent and isolated systems

- SIGACAD
- SIGEF
- RHumo
- PACO and RIA platforms

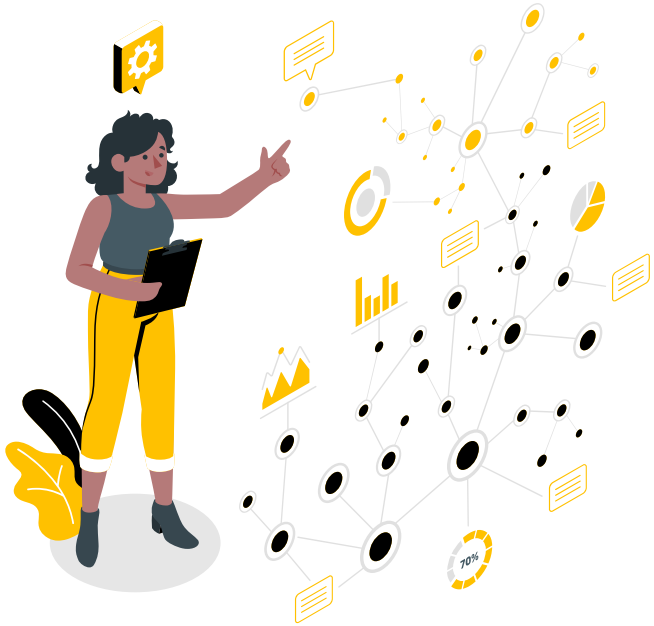
Data

Heterogeneous and fragmented data sources

No centralized access to data

BI & Solution

03



How BI integrates data and supports decision-making

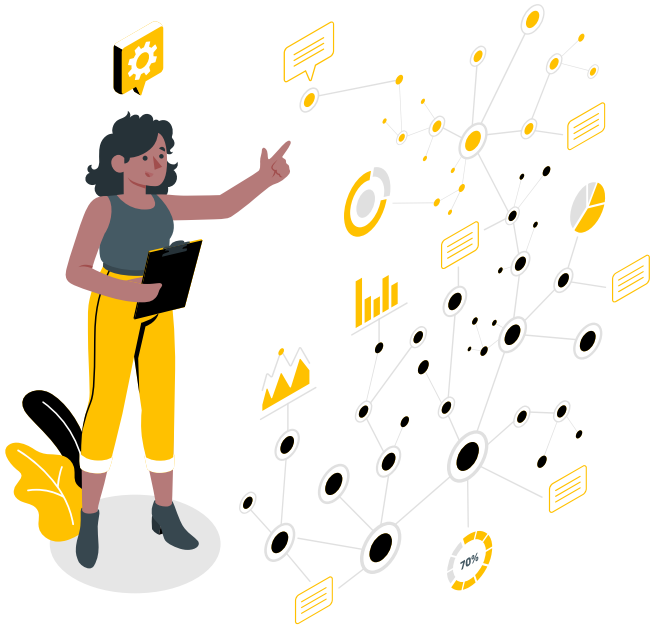
BI & Solution

What Business Intelligence provides:

- Data integration across all systems
- Centralized and unified data view
- Data → Information → Insights
- Improved and faster decision-making

Architecture, Functionalities and Requisites

04



How the BI system works and what it must provide

4.1 Architecture

ETL PIPELINE

Data flows from heterogeneous sources → ETL processing → structured Data Warehouse → BI tools deliver insights to decision-makers.

4.2 Functionalities

Dashboards

Interactive and dynamic visualizations of KPIs and metrics

KPIs

Definition and monitoring of key performance indicators

Reporting

Ad-hoc and periodic report generation

Interactive Analysis

Drill-down and cross-filter data exploration

Data Integration

ETL processes from multiple heterogeneous sources

Visual Analytics

40+ chart types: graphs, maps, pivot tables

4.3 Requirements

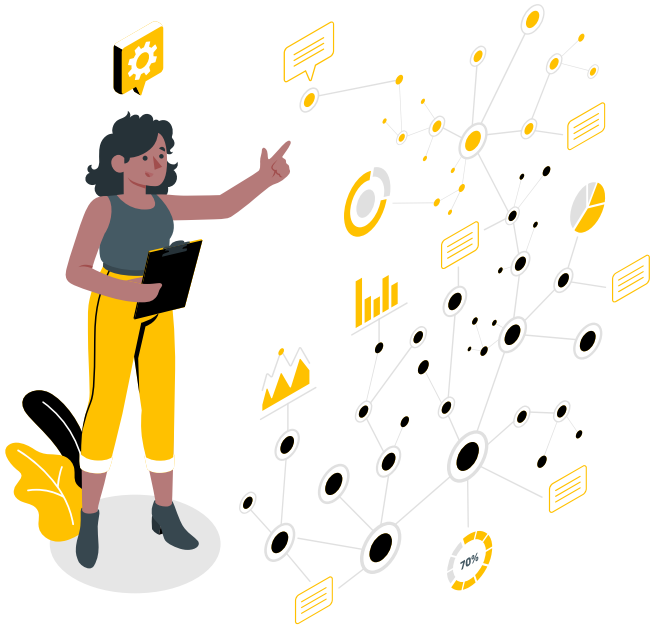
Functional Requirements

Requirement	Description
Interactive Dashboards	Dynamic and intuitive dashboards
KPIs	Definition & monitoring of performance indicators
Reporting	Ad-hoc and periodic reports
Interactive Analysis	Explore data in detail
Data Integration	Combine data from different sources

Non-Functional Requirements

Requirement	Description
Multi-source support	Connect to different databases
SQL Support	Execute SQL queries
Scalability	Handle large data volumes
Authentication	User access control
Authorization	Permission management by profile
Intuitive Interface	Easy to use for all user types

Market Analysis



05

Overview and comparison of available BI tools

5.1 Platforms Identified

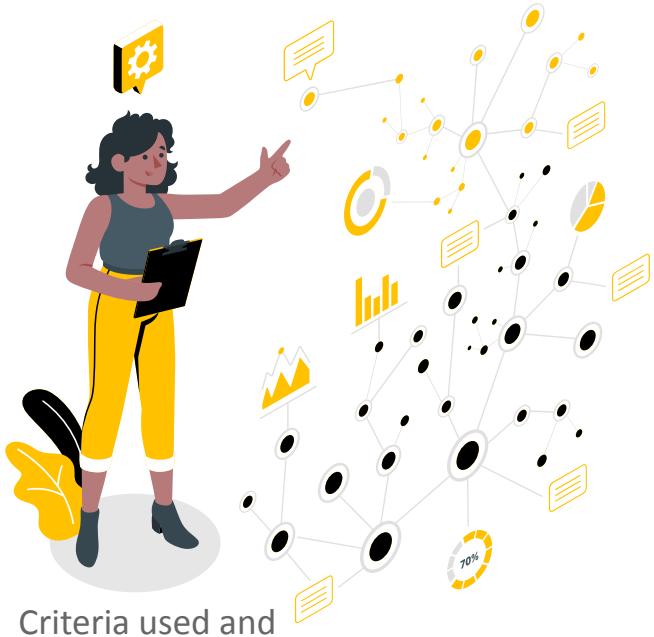
Platform	Type	SQL	Viz	Scale	Cost	Highlights
Metabase	Open Source	✓	✓	Medium	Free	Simple UI, 20+ DB sources
Apache Superset	Open Source	✓	✓	High	Free	40+ chart types, scalable
KNIME	Open Source	✓	✓	High	Free	300+ connectors, ETL + ML
Lightdash	Open Source	✓	✓	Medium	Free	dbt integration, NLP
Redash	Open Source	✓	✓	Medium	Free	SQL-focused, NoSQL support
Looker Studio	Free	✓	✓	High	Free	800+ connectors, Google
Power BI Free	Freemium	✓	✓	Medium	Limited	Microsoft 365 integration
Tableau Public	Freemium	Lim.	✓	Low	Limited	High-quality visualizations
Jaspersoft CE	Open Source	✓	✓	Medium	Free	Robust reporting engine
BIRT	Open Source	✓	✓	Medium	Free	Java integration

5.2 Selection Criteria

Criterion	Weight	Justification
Multi-source Integration	**High**	UA has SIGACAD, SIGEF, RHumo, PACO, RIA
SQL Support	**High**	Data Warehouse will be relational
Visualization Capability	**High**	Dashboards and KPIs are central requirements
Ease of Use	**Medium**	Accessible to all user profiles at UA
Scalability	**Medium**	Growing academic, financial and HR data
Cost (Open Source)	**High**	Public university, preference for free tools
Security & Access Control	**High**	Sensitive student, staff and financial data
Community & Support	**Medium**	Facilitate maintenance and system evolution

Selection & Decision

06



Criteria used and final tool selection

Apache Superset — Selected Platform

Why Apache Superset?

- Native SQL via SQLAlchemy — fully compatible with UA's Data Warehouse
- 40+ visualization types: charts, maps, pivot tables
- No-code builder + SQL IDE — suitable for all user profiles
- Highly scalable for growing academic, HR and financial data
- 100% Open Source under Apache Software Foundation — zero licensing cost
- Role-based authentication and access control
- Cross-filter, drill-by and caching for deep interactive analysis

Evaluation Summary

SQL Integration	■■■■■
Visualization	■■■■■
Scalability	■■■■■
Ease of Use	■■■■□
Cost	■■■■■
Security	■■■■□
Community	■■■■■

✓ **Most balanced open-source BI solution for UA**

Thanks!

Do you have any questions?